

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress

Go Back

Back To Practice

Round Robin CPU Scheduling Algorithm

Write a C program to implement the Round Robin (RR) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), Burst Times (BT), and a Time Quantum (TQ). The CPU should rotate through the processes in the ready queue, granting each a maximum of one TQ of execution time. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and the overall averages.

Input Format

1. The first line of input contains an integer representing the number of processes.

2. The second line contains an integer representing the Time Quantum.

3. The next lines contain two integers for each process, where

- The first integer denotes the Arrival Time (AT)
- The second integer denotes the Burst Time (BT) of the process.

Output Format

The output displays the Average Turnaround Time and the Average

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, tq, i;
6
7     printf("Enter number of processes: \n");
8     scanf("%d", &n);
9
10    printf("Enter Time Quantum: \n");
11    scanf("%d", &tq);
12
13    int AT[n], BT[n], RT[n];
14    int CT[n], TAT[n], WT[n];
15    int completed = 0;
16
17    for(i = 0; i < n; i++)
18    {
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

Custom Test

Sample Test Case

Test Case - 1

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress

Go Back

Back To Practice

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Output Format

The output displays the Average Turnaround Time and the Average

Select Language: C (GCC 9.2.0)

```
19    printf("Enter AT and BT for P%d: \n", i + 1);
20    scanf("%d %d", &AT[i], &BT[i]);
21    RT[i] = BT[i];
22 }
23
24 int current_time = 0;
25
26 while(completed < n)
27 {
28     int executed = 0;
29
30     for(i = 0; i < n; i++)
31     {
32         if(AT[i] <= current_time && RT[i] > 0)
33         {
34             executed = 1;
35
36             if(RT[i] > tq)
37             {
```

Test Cases 2

Run Sample Cases

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Custom Test

Sample Test Case

Test Case - 1

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress

Go Back

Back To Practice

Sample Input 1

3
2
0 5
1 3
2 1

Sample Output 1

Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
Average Waiting Time: 3.33

Select Language: C (GCC 9.2.0)

```
38         current_time += tq;  
39         RT[i] -= tq;  
40     }  
41     else  
42     {  
43         current_time += RT[i];  
44         RT[i] = 0;  
45  
46         CT[i] = current_time;  
47         completed++;  
48     }  
49 }  
50  
51  
52     if(executed == 0)  
53         current_time++;  
54 }  
55  
56 float totalTAT = 0, totalWT = 0;
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress

Go Back

Back To Practice

Sample Input 1

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2
0 5
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2 1

Sample Output 1

Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
Average Waiting Time: 3.33

Select Language: C (GCC 9.2.0)

```
57  
58     for(i = 0; i < n; i++)  
59     {  
60         TAT[i] = CT[i] - AT[i];  
61         WT[i] = TAT[i] - BT[i];  
62  
63         totalTAT += TAT[i];  
64         totalWT += WT[i];  
65     }  
66  
67     float avgTAT = totalTAT / n;  
68     float avgWT = totalWT / n;  
69  
70     printf("\nAverage Turnaround Time: %.2f\n", avgTAT);  
71     printf("Average Waiting Time: %.2f\n", avgWT);  
72  
73     return 0;  
74 }
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1